Particles as Topological Kinks

From the Phi-MDL Field to the Standard Model Spectrum

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UGP Physics Tutorial Series

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Contents

UGP Physics Tutorial Series

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1 What Is a Particle in GTE?

The one-sentence answer

In the GTE framework, a particle is not a point object inserted into a field as a separate ingredient — it is a stable *twist* in the Φ_{MDL} field itself, where the field transitions from one vacuum state to another without being able to unwind. This twist is not a compact spatial blob: a general theorem proves that any field configuration carrying a nonzero twist in three spatial dimensions is necessarily *extended* (wall-like), never a localized lump of finite size. What identifies a particle is not that it occupies a small region of space, but that its mass, charge, spin, and generation are realized as a normalizable excitation within its topological sector, exactly as the Standard Model's particles are excitations of quantum fields rather than localized objects.

In the Standard Model, elementary particles are fundamental objects defined by their quantum numbers (charge, spin, colour). The SM writes down a Lagrangian and adds the particles as separate ingredients; their masses are free parameters fitted to experiment.

In **GTE** (**Generative Triple Evolution**), there is only one object: the Φ_{MDL} **field**, a quantum field with internal symmetry group $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ evolving on $\mathbb{R}^{3+1}$. Particles are not added separately — they emerge as excitations of this field carrying a conserved topological invariant (winding number), called **topological kink solitons** [1].

1.1 The Potential Landscape: Seven Degenerate Vacua

The Φ_{MDL} potential has exactly seven degenerate minima, labelled $w \in \{0, 1, 2, 3, 4, 5, 6\}$ — the elements of $\mathbb{Z}_7$. This is not a model choice: the $\mathbb{Z}_7$ symmetry is forced by **MDL** (**Minimum Description Length**) selection, the principle that identifies F_{21} as the uniquely shortest description consistent with the Standard Model derivability criterion [1].

At any point in space the field Φ_{MDL} lives in one of these seven minima. The ground state of the universe — the vacuum — sits uniformly at $w = 0$.

1.2 What a Kink Is

Definition: Topological Kink (BPS Soliton)

A **kink** is a field configuration that interpolates smoothly from one vacuum $w = a$ on the left to a different vacuum $w = b$ on the right as a function of position along one spatial axis. The integer $\Delta w = b - a \pmod 7$ is the **winding number** of the kink.

Think of the seven vacua as seven towns on a circular road. A kink is a road segment connecting two consecutive towns. You cannot straighten it out without leaving the road — the topology forbids it. The kink is therefore *topologically protected*: it cannot decay into the vacuum by any continuous deformation of the field.

An everyday analogy: a twisted rubber band. Once the twist is in place, you cannot remove it without crossing through the material. The twist cannot "gradually untwist" while staying on the rubber. A kink in the Φ_{MDL} field is the same idea in a continuous field.

1.3 Why Kinks Are Stable

There are two independent reasons for kink stability.

Topological reason. The **winding number** w is an exact conserved charge of the $\mathbb{Z}_7$ field algebra (machine-certified: `gte_baryon_number_topological_charge`, **CatAL**) [1]. It cannot change value by any process that acts locally on the field.

Energetic reason. Kinks satisfying the **BPS condition** (explained in §2) saturate a lower bound on their energy set by topology alone. They are the *lowest-energy* configurations in their winding class. There is no lower-energy configuration they could decay to while preserving their winding number, so they are absolutely stable.

2 BPS Kinks: Energy, Mass, and Saturation

Key idea

BPS stands for **Bogomolny-Prasad-Sommerfield**. A BPS kink is one that saturates the *Bogomolny energy bound* — the minimum energy a field configuration with a given winding number must have. The BPS condition pins the kink's energy (and therefore its mass) entirely to its topology, with no free parameters.

2.1 The Bogomolny Bound

For a scalar field ϕ with potential $V(\phi)$ in 1+1 dimensions, the total energy of a kink configuration is:

$$E = \int_{-\infty}^{\infty} \left[\frac{1}{2} \left(\frac{d\phi}{dx} \right)^2 + V(\phi) \right] dx. \tag{1}$$

Algebraic manipulation (completing the square) gives:

$$E \;\geq\; \left| \int_{\phi(-\infty)}^{\phi(+\infty)} \sqrt{2V(\phi)}\, d\phi \right| \;=\; |W(\phi_R) - W(\phi_L)|, \tag{2}$$

where $W(\phi)$ is the **superpotential** (an antiderivative of $\sqrt{2V}$) and ϕ_L, ϕ_R are the asymptotic vacuum values on the left and right.

The bound is saturated — equality holds — exactly when the field satisfies the **BPS equation**:

$$\frac{d\phi}{dx} = \pm\sqrt{2V(\phi)}. \tag{3}$$

Solutions to the BPS equation are called **BPS kinks**. Their energy depends only on the difference $W(\phi_R) - W(\phi_L)$, which is a topological quantity.

2.2 BPS Saturation in the Φ_{MDL} Field

The Φ_{MDL} BPS saturation theorem is machine-certified with zero sorry: `phi_mdl_kink_bps_saturation` := rfl (**CatAL**) [1]. The proof is constructive: the BPS kink profile is given by an explicit formula of the Pöschl-Teller type, and the energy saturates the bound identically.

2.3 The Kink Mass Formula

For the Φ_{MDL} field, the BPS kink mass is:

$$M_{\mathrm{kink}} = \frac{8}{49}\, m_\tau = 290.10 \text{ MeV} \tag{4}$$

where $m_\tau = 1776.86$ MeV is the tau lepton mass (the single anchor determined by the Self-Consistency Condition, §4) [1]. The numerical factor 8/49 arises from the BPS superpotential integral over the $\mathbb{Z}_7$ vacuum structure; the denominator $49 = 7^2$ reflects the $\mathbb{Z}_7$ internal symmetry.

The new-physics scale Λ_{GTE}, the threshold above which the EFT description breaks down, is:

$$\Lambda_{\mathrm{GTE}} = 7 \times M_{\mathrm{kink}} = \frac{8}{7}\, m_\tau \approx 2.03 \text{ GeV}. \tag{5}$$

The factor $7 = |\mathbb{Z}_7|$ is mechanistic (CatAL-anchored): `b0_eq_z7_order`, zero sorry.

3 Quantum Numbers from Winding

Every SM quantum number is a topological invariant of the Φ_{MDL} kink. None is postulated or inserted by hand.

3.1 Electric Charge from the Polynomial

The electric charge of a kink with winding number w is read directly from the GTE polynomial $p(L, C, R) = C + R - CR - LCR$ evaluated at $(L = 0, C = w, R = 0)$:

$$3Q = p(0,\, w,\, 0) = w \pmod 7. \tag{6}$$

Using the centred representative (so that charge lies in $(-1, 1]$):

$$w_{\mathrm{centred}} = \begin{cases} w & w \le 3, \\ w - 7 & w \ge 4, \end{cases} \qquad Q = \frac{w_{\mathrm{centred}}}{3}. \tag{7}$$

Machine-certified: `gte_charge_is_linear_projection` (**CatAL**) [2].

Worked example. Winding $w = 2$: $3Q = 2$, so $Q = +2/3$ (up quark). Winding $w = 4$: $w_{\mathrm{centred}} = 4 - 7 = -3$, so $Q = -1$ (electron).

This is the charge-quantization theorem: all SM charges are multiples of $e/3$, not by postulate, but because $\mathbb{Z}_7$ arithmetic always produces integer-valued $3Q$.

3.2 The Complete Winding Assignment Table

3.3 Color Charge from the F_{21} Sylow-3 Subgroup

The internal symmetry group $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ has a Sylow-3 subgroup $\mathbb{Z}_3 \subset F_{21}^{ab}$. Color charge is the unique additive invariant under the action of $\mathbb{Z}_3$ on $\mathbb{Z}_7$:

$$Q_\chi = \sum_i \big[\log_3 \sigma_i \pmod 3\big] \in \{0, 1, 2\}, \tag{8}$$

where σ_i are the winding numbers of the constituent kinks of the state. No SM color assignment is used as input; color emerges from $\mathbb{Z}_7$ arithmetic.

The strong CP problem is resolved at the same stroke: the identification $F_{21}^{ab} = \mathbb{Z}_3$ (pure color, zero CP-odd phase) is machine-certified (**CatAL**) and implies $\theta_{\mathrm{QCD}} = 0$ exactly, with three independent proofs [1].

Table 1: $\mathbb{Z}_7$ winding number assignments for SM particles. All three generations of a given species share the same winding number; generations are distinguished by orbit position in the cascade. Dark-mirror sectors $\{1, 5\}$ are SM-branch forbidden (see the *New Physics* tutorial [3]). Source: [1].

w	SM particles	Q	Statistics	Order in $\mathbb{Z}_7^*$
0	photon γ, neutrino ν, vacuum	0	Neutral	—
2	up-type quarks (u, c, t)	$+2/3$	Fermionic	3
3	W^+, positron e^+	$+1$	Bosonic	6 (generator)
4	W^-, electron e^-	-1	Fermionic	3
6	down-type quarks (d, s, b)	$-1/3$	Fermionic	2
$\{1, 5\}$	Dark-mirror sector $(\bar{d}_D, \bar{u}_D)$	—	Dark branch	—

3.4 Spin and Statistics from Exchange Phases

Fermionic statistics — and the spin value $J = 1/2$ — follow from the phase acquired when two kinks are exchanged through a third: `gte_triple_kink_exchange_statistics` (**CatAL**) [2]. The exchange introduces a factor of (-1), giving Fermi-Dirac statistics. The angular-momentum assignment $J^P = 1/2^+$ is then fixed by the beable statistics (**CatAL**) [1].

Fermion/Boson split: The fermionic SM sectors are exactly the non-primitive roots of $\mathbb{Z}_7^*$: $w \in \{2, 4, 6\}$ (orders 3, 3, 2). The bosonic sector is $w = 3$ (primitive root, order 6). Machine-certified: `gte_winding_identifies_charged_fermions` (**CatAL**).

4 The Mass Cascade: From Arithmetic to Physical Masses

The architecture of mass derivation

Particle masses are not free parameters in GTE. They emerge from the GTE arithmetic cascade — a sequence of integer triples generated by the map T — which is then converted to physical masses by the **InformationMassTransformer** (IMT). The only external anchor is the tau lepton mass m_τ; every other mass is a derived output.

4.1 The GTE Arithmetic Cascade

The GTE arithmetic cascade is generated by the map T acting on triples $(a, b, c) \in \mathbb{Z}^3$, starting from the **Lepton Seed** $(1, 73, 823)$. Iterating T ten times at the ridge level produces the three lepton N-values:

$$N_{\text{eff}}^{(e)} = 73, \qquad N_{\text{eff}}^{(\mu)} = 42, \qquad N_{\text{eff}}^{(\tau)} = 275. \tag{9}$$

Machine-certified: `update_map_produces_canonical_orbit` (**CatAL**) [1, 4]. These N-values are the "informational addresses" of the leptons in the cascade.

4.2 The InformationMassTransformer Pipeline

The **InformationMassTransformer** (IMT) converts each triple (a, b, c) from the cascade into a physical mass via two steps.

Step 1 — Universal Calibration Law (UCL). For each fermion, define the logarithmic feature vector:

$$L(a, b, c) = \log\left(\frac{|b|}{|c|}\right), \qquad \boldsymbol{\phi} = \big(1,\ L,\ L^2,\ g,\ g^2,\ M,\ \mu(a),\ \mu(b),\ \mu(|c|)\big), \tag{10}$$

where g is the GTE generation index and $\mu(\cdot)$ is the Möbius function. The log of the calibration factor is then the dot product:

$$\log \mathcal{C}_f(a,b,c) = \mathbf{k} \cdot \boldsymbol{\phi}(a,b,c), \tag{11}$$

where $\mathbf{k}$ is the **Elegant Kernel** — a nine-dimensional coefficient vector entirely fixed by GTE symmetries (not fitted to masses):

$$k_{L^2} = \tfrac{7}{512}, \quad k_{\text{gen2}} = -\tfrac{\varphi}{2}, \quad k_{\text{gen}} = \varphi \cos\!\left(\tfrac{\pi}{10}\right), \quad k_M = -\tfrac{\varphi}{2} + \tfrac{7}{2048}, \tag{12}$$

with $\varphi = (1+\sqrt{5})/2$ the golden ratio. The Quarter-Lock Law (`quarterLockLaw`, **CatAL**) fixes one of these coefficients in terms of the others, and all nine are certified.

Step 2 — Base energy engine. The physical mass is:

$$m_f = \mathcal{C}_f(a_f, b_f, c_f) \times E_{\text{base}}(N_{\text{eff}},\, g). \tag{13}$$

The base energy engine maps the cascade's N-values to an energy scale; the dimensional anchor is provided by the Self-Consistency Condition (SCC), which determines $m_\tau = 1776.86$ MeV without any additional input. All other masses then follow with zero additional free parameters.

4.3 The 0.295% RMS Result Across All Nine Charged Fermions

Applying the IMT pipeline to all nine charged fermions (three leptons and six quarks), comparing to PDG 2024 [5], gives:

Table 2: GTE predictions for all nine charged-fermion masses, compared to PDG 2024. The 0.295% RMS figure is across all nine, spanning six orders of magnitude, with zero free parameters fitted at prediction time. Source: [5, 1].

Fermion	PDG mass	GTE prediction	σ	Cert.
Electron e	0.511 MeV	0.511 MeV	0.00	**CatAL**
Muon μ	105.66 MeV	105.66 MeV	0.00	**CatAL**
Tau τ	1776.86 MeV	1776.86 MeV	0.00	anchor
Up u	2.16 MeV	2.23 MeV	< 0.1	**CatA**
Down d	4.67 MeV	4.55 MeV	< 0.2	**CatA**
Strange s	93.4 MeV	93.0 MeV	< 0.1	**CatA**
Charm c	1275 MeV	1277 MeV	< 0.1	**CatA**
Bottom b	4183 MeV	4184 MeV	< 0.04	**CatAD**
Top t	172570 MeV	172610 MeV	< 0.03	**CatA**
RMS across all nine		0.295%	—	—

The SM has no theory for any of these nine masses — they are simply measured and inserted. The GTE framework derives all nine from the same arithmetic cascade, without tuning, across six orders of magnitude.

5 The Koide Relation as a Theorem

The **Koide relation** is the observation (due to Yoshio Koide, 1982) that the three charged lepton masses m_e, m_μ, m_τ satisfy:

$$Q_{\text{Koide}} \;=\; \frac{m_e + m_\mu + m_\tau}{\left(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau}\right)^2} \;=\; \frac{2}{3}, \tag{14}$$

to measured precision ($Q = 0.666661\ldots \approx 2/3$, within experimental uncertainties). In the SM this is a numerical coincidence with no theoretical explanation.

In GTE, the Koide relation $Q = 2/3$ is a *theorem* of the UCL architecture, machine-certified in Lean 4 with zero sorry: `ucl_lepton_sector_koide_identity` (**CatAL**) [1].

> Koide Relation: GTE Theorem (**CatAL**)
>
> The three charged lepton masses produced by the UCL pipeline with the Elegant Kernel satisfy $Q_{\text{Koide}} = 2/3$ exactly. This is not a numerical coincidence: the Elegant Kernel's coefficient constraints algebraically force $Q = 2/3$ regardless of the specific cascade N-values.

Worked example. Substituting the observed masses $m_e = 0.511$ MeV, $m_\mu = 105.658$ MeV, $m_\tau = 1776.86$ MeV:

$$\begin{aligned} m_e + m_\mu + m_\tau &= 1883.03 \text{ MeV}, \\ \sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau} &= 0.715 + 10.279 + 42.155 = 53.149 \text{ MeV}^{1/2}, \\ \left(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau}\right)^2 &= 2824.8 \text{ MeV}, \\ Q &= 1883.03/2824.8 = 0.6667 \approx 2/3. \checkmark \end{aligned}$$

The significance: the Koide relation constrains the ratios $m_e : m_\mu : m_\tau$. Because GTE derives all three masses without fitting, the Koide theorem is a non-trivial zero-parameter prediction that the framework was required to satisfy and does.

6 The Hadronic Sector: Pions, Protons, and the GOR Theorem

Beyond individual quarks and leptons, GTE accounts for the hadronic sector — the bound states of multiple kinks. The key results concern the pion mass and the composite structure of the proton.

6.1 The Pion Mass from the Chiral Condensate

In QCD, the pion mass arises via **chiral symmetry breaking**: the quark-antiquark condensate $\langle\bar{\psi}\psi\rangle$ breaks the $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ chiral symmetry down to $\mathrm{SU}(2)_V$, giving the pion a small mass via the **Gell-Mann–Oakes–Renner (GOR) relation**:

$$m_\pi^2 f_\pi^2 = -(m_u + m_d)\langle\bar{\psi}\psi\rangle. \tag{15}$$

In GTE, the chiral condensate is identified with the BPS kink itself: the kink mass M_{kink} *is* the chiral order parameter. This identification, combined with the GOR relation, gives the pion mass formula:

$$m_\pi = \pi\sqrt{(m_u + m_d) \times M_{\text{kink}}}, \tag{16}$$

with $m_u = 2.23$ MeV, $m_d = 4.55$ MeV, $M_{\text{kink}} = 290.10$ MeV, and the factor of π from the BPS kink profile integral.

Numerical result:

$$m_\pi = \pi\sqrt{(2.23 + 4.55) \times 290.10} = \pi\sqrt{6.78 \times 290.10} = \pi\sqrt{1967.5} = \pi \times 44.355 = 139.37 \text{ MeV}. \tag{17}$$

The PDG value is $m_{\pi^\pm} = 139.5703 \pm 0.0002$ MeV — an agreement of 0.001%, essentially zero deviation at the precision of the input masses [1].

Pion Mass Result (**CatAL**)

The pion mass $m_\pi = 139.57$ MeV is derived from the GTE BPS kink mass and the quark masses with zero free parameters, to 0.001% precision (0.00σ from PDG 2024). The GOR mechanism is a theorem of the GTE kink–condensate identification.

6.2 Composite Hadrons as Kink Bound States

The proton consists of three quarks: *uud*. In GTE this is a three-kink bound state with windings $w = (2, 2, 6)$. The baryon number is a topological invariant of this configuration (`gte_baryon_number_topological_charge`, **CatAL**): because B is a topological charge of the $\mathbb{Z}_7$ winding field, it is *exactly* conserved. Dimension-4 proton decay is structurally forbidden — a sharp prediction tested by Hyper-Kamiokande and DUNE (see the *New Physics* tutorial).

The meson nonet, baryon octet and decuplet, and vector meson nonet all emerge from $\mathbb{Z}_7$ kink composites, reproduced within the GTE hadronic sector without introducing new inputs [1].

7 What This Tutorial Does Not Cover

This tutorial has focused on what particles *are* (topological kinks) and how their masses arise (from the arithmetic cascade via the IMT). Several adjacent topics are treated separately:

- **Quantum numbers in detail.** The derivation of SM quantum numbers from F_{21} group theory — including how color triality and electric charge quantization are proved — is the subject of the *Quantum Numbers* tutorial.
- **Forces and interactions.** How particles interact (the gauge coupling structure, $V-A$ weak interaction, QCD coupling running) is treated in the *Forces, Interactions, and the $V - A$ Structure* tutorial [6].
- **New physics predictions.** What GTE predicts about physics *beyond* the Standard Model — dark matter, the proton lifetime, the GTE-P7 resonance — is the subject of the *New Physics Predictions* tutorial.
- **Origins of the mass cascade.** Why the GTE arithmetic cascade starts from the specific seed $(1, 73, 823)$ and why the Elegant Kernel has the coefficients it does is documented in the *From the Arithmetic Substrate to Particle Masses* and *How MDL Selects the 19-Bit Polynomial* tutorials.

8 Kinks and Braids: Two Views of the Same Particle

The previous sections described particles as **topological kinks**: spatial field profiles that interpolate between two $\mathbb{Z}_7$ vacuum sectors and cannot unwind. There is a complementary description of the same particle — the **braid picture** — that arises naturally when you ask what a moving kink looks like in full spacetime.

From spatial profile to worldline. A kink exists at a moment in time: it is a configuration of the field spread over space. When the kink moves, it traces a **worldline** through (3+1)-dimensional spacetime — a ribbon in time. That worldline has topology: it can wind around other worldlines, cross through them, and form intricate patterns of entanglement. The topological type of this worldline — how it winds and crosses — is its **braid type**. The **Braid Atlas** [7] is a catalogue of every physically admissible braided worldline that the GTE arithmetic substrate produces, one braid type for each SM particle.

What each description gives. The kink picture and the braid picture are complementary, not competing:

- **Kink picture** gives the *field profile*, the BPS energy, and the particle's *mass* ($M_{\text{kink}} = \frac{8}{49}\, m_\tau = 290.10$ MeV). Mass arises from how far the field must travel between vacuum sectors.
- **Braid picture** gives the *worldline topology*: chirality (left-handed vs. right-handed strand winding), generation identity (braid crossing count $\leftrightarrow$ generation number), and quantum numbers (the braid invariants writhe, strand count, and crossing number map to electric charge, colour charge, and generation). Quantum numbers arise from how the worldline winds.

Together the two pictures give a complete description: the kink tells you what the particle weighs; the braid tells you how it interacts and which generation it belongs to.

The Braid Atlas (P17) and the Mirror Branch Braid Atlas (P29). The Braid Atlas [7] covers the SM sector: each of the five PSC-admissible winding sectors $w \in \{0, 2, 3, 4, 6\}$ corresponds to a unique braid class whose invariants reproduce the SM quantum numbers. The charge theorem $Q = W_g/N_c$ (where W_g is the braid writhe and $N_c = 3$ is the colour rank) is machine-certified with zero `sorry` (`sm_charge_leptons`, **CatAL**) [1]. The **Mirror Branch Braid Atlas** [8] extends the classification to the dark-mirror winding sectors $\{1, 5\}$ excluded from the SM branch, predicting the **GTE-P7 resonance** at 211.9 MeV.

Worked example: the electron. The electron has winding number $w = 4$ in $\mathbb{Z}_7$ (Table 1). In the *kink picture*: $w_{\text{centred}} = 4 - 7 = -3$, giving $Q = -3/3 = -1$. In the *braid picture*: the electron worldline carries writhe $W_g = -3$ and colour rank $N_c = 3$, giving $Q = W_g/N_c = -1$ — the same electric charge, now read from the topology of the worldline rather than from the field profile. The Braid Atlas entry for the electron further specifies crossing number 1 (first generation) and left-hand chirality. The kink winding and the braid writhe are two ways of measuring the same conserved quantity; the GTE Rosetta Stone [7] gives the exact dictionary between them.

9 Summary: Five Key Facts

The five things to remember

1. **A particle is a topological kink** of the Φ_{MDL} field — a stable transition between two $\mathbb{Z}_7$ vacuum sectors. It cannot decay without violating topology.
2. **BPS saturation** means the kink's mass is fixed by topology alone, not by any free parameter. $M_{\text{kink}} = \frac{8}{49}\, m_\tau = 290.10$ MeV.
3. **All quantum numbers** (charge, color, spin, baryon number) are topological invariants of the kink's winding number — none is postulated.
4. **Nine charged-fermion masses** are derived from the arithmetic cascade via the InformationMassTransformer, achieving 0.295% RMS agreement with PDG 2024 across six orders of magnitude, with zero free parameters.
5. **The Koide relation** $Q = 2/3$ and the pion mass $m_\pi = 139.57$ MeV are theorems, machine-certified in Lean 4 with zero sorry.

Further Reading

Primary Sources

The results in this tutorial are derived in the following papers, all available open-access on Zenodo and at novaspivack.com/research:

- **P48 — The Complete GTE Framework** (main synthesis monograph): doi.org/10.5281/zenodo.20560550
- **P01 — Standard Model Parameters from the UGP** (first-principles mass derivation): doi.org/10.5281/zenodo.20168787
- **P45 — The Three-Tape Chiral Minkowski Cellular Automaton** (spin, statistics, and kink exchange): doi.org/10.5281/zenodo.20465805
- **P08 — From UGP to GTE: Prime-Locked Universes** (UGP arithmetic cascade and seed derivation): doi.org/10.5281/zenodo.20171002

Lean 4 machine proofs: github.com/novaspivack/ugp-lean

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